The University of Azad Jammu and Kashmir, Muzaffarabad

Course: Computer Network

Course Code: CS-2203

Roll No: 2024-SE-13

Lab: 1& 2

Submitted To: Engr. Asma Javaid

Submitted By: Sughra Mumshad

Date: 11 March, 2026

Department Of Software Engineering

Part 1: Installation of Cisco Packet Tracer

Lab Title

Designing and Simulating a Basic Network using Cisco Packet Tracer

1. Objective

- To understand the basic concepts of computer networking.
- To learn how to use Cisco Packet Tracer.
- To design and simulate a simple LAN (Local Area Network).
- To configure IP addresses and test connectivity using ping command.

2. Introduction

Cisco Packet Tracer is a network simulation software developed by Cisco Systems. It allows students to create network topologies, configure devices, and test communication between them without using real hardware.

It is widely used for learning networking concepts such as:

- LAN and WAN design
- IP addressing
- Router and switch configuration
- Network troubleshooting

3. Software Requirements

- Cisco Packet Tracer (Installed)
- Windows 10/11 or any compatible OS
- Basic knowledge of IP addressing

4. Theory

What is a LAN?

A LAN (Local Area Network) connects computers within a small area like a lab, office, or home.

Devices Used in This Lab:

- 2 PCs
- 1 Switch
- Copper Straight-Through Cable

IP Address

An IP address uniquely identifies a device in a network.
Example: 192.168.1.1

5. Procedure

Step 1: Open Cisco Packet Tracer

Open the software.

Step 2: Add Devices

1. Click on **End Devices**
2. Select **PC** and drag two PCs into workspace.
3. Click on **Switches**
4. Select **2960 Switch** and drag into workspace.

Step 3: Connect Devices

1. Click on **Connections**
2. Select **Copper Straight-Through**
3. Connect:
    - PC0 → Switch (FastEthernet0/1)
    - PC1 → Switch (FastEthernet0/2)

Step 4: Configure IP Addresses

For PC0:

- Click PC0 → Desktop → IP Configuration
- IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0

For PC1:

- Click PC1 → Desktop → IP Configuration
- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

Step 5: Test Connectivity

1. Click PC0
2. Go to Desktop → Command Prompt
3. Type:

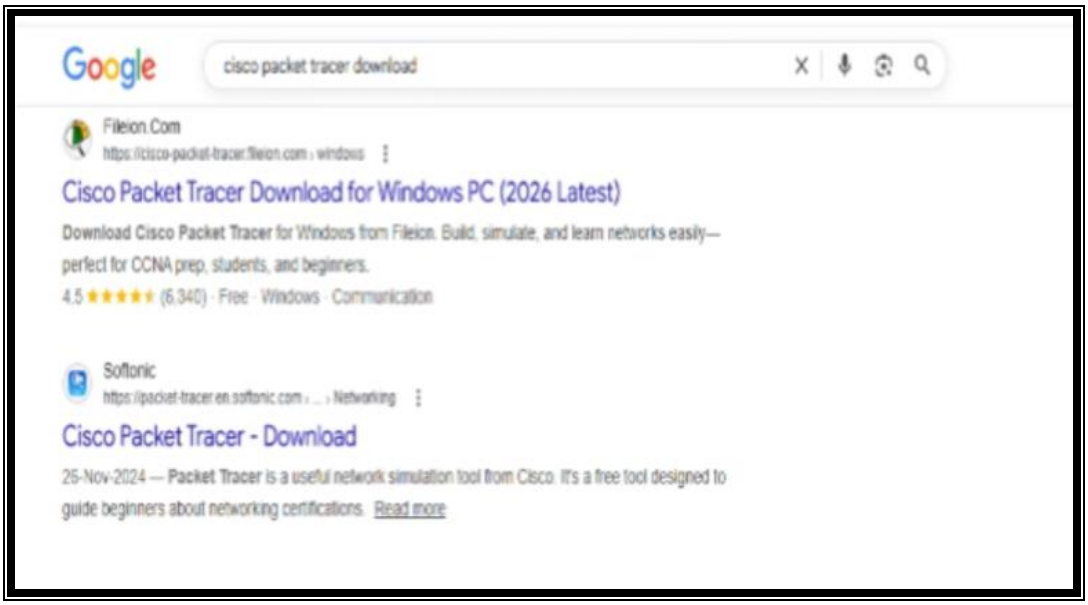
```
ping 192.168.1.2
```

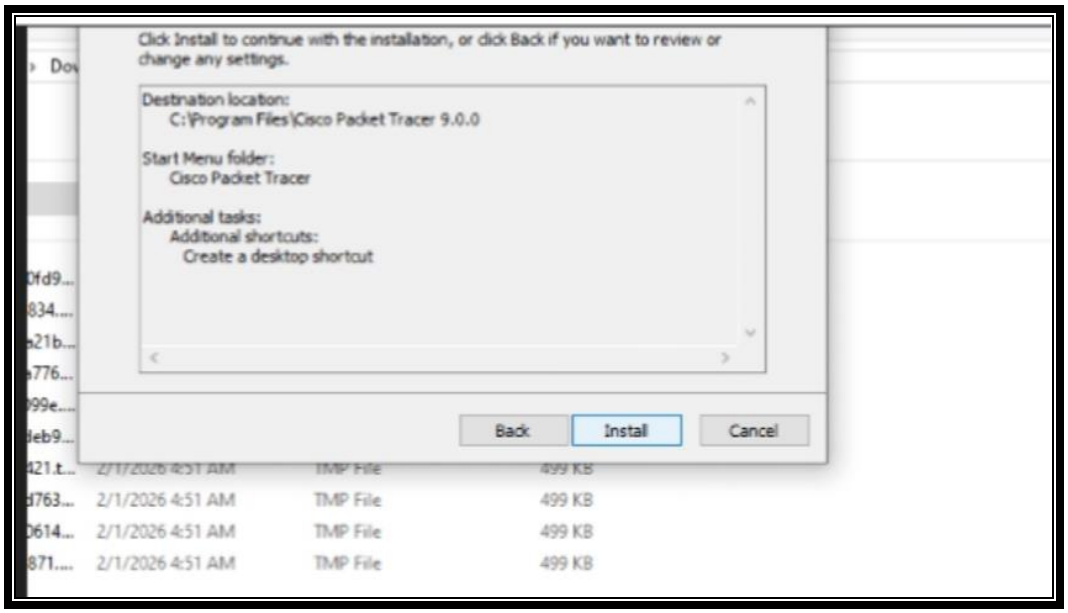
If configuration is correct, you will see:

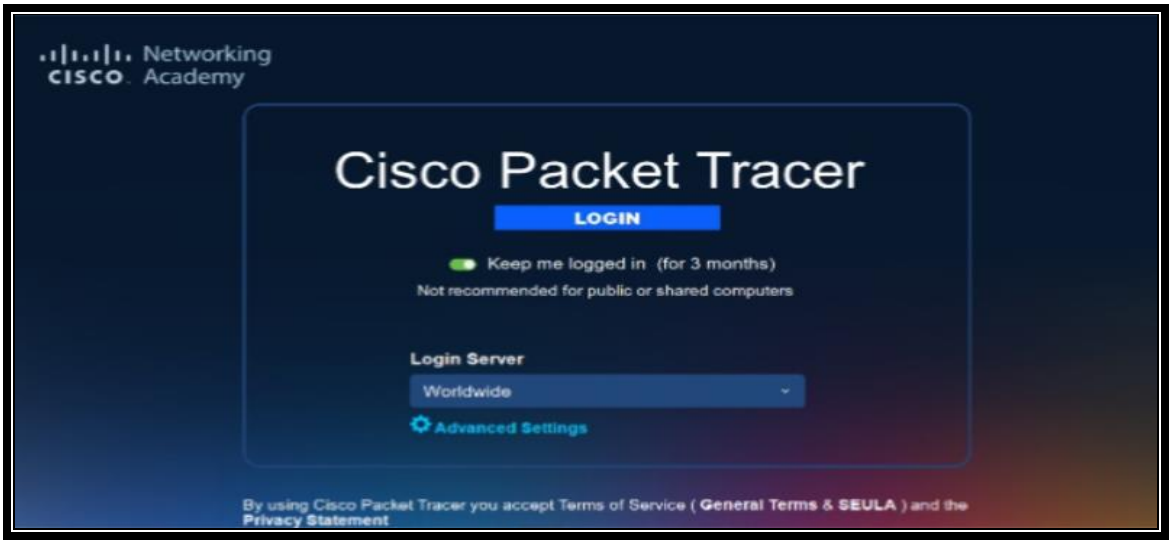
```
Reply from 192.168.1.2
```

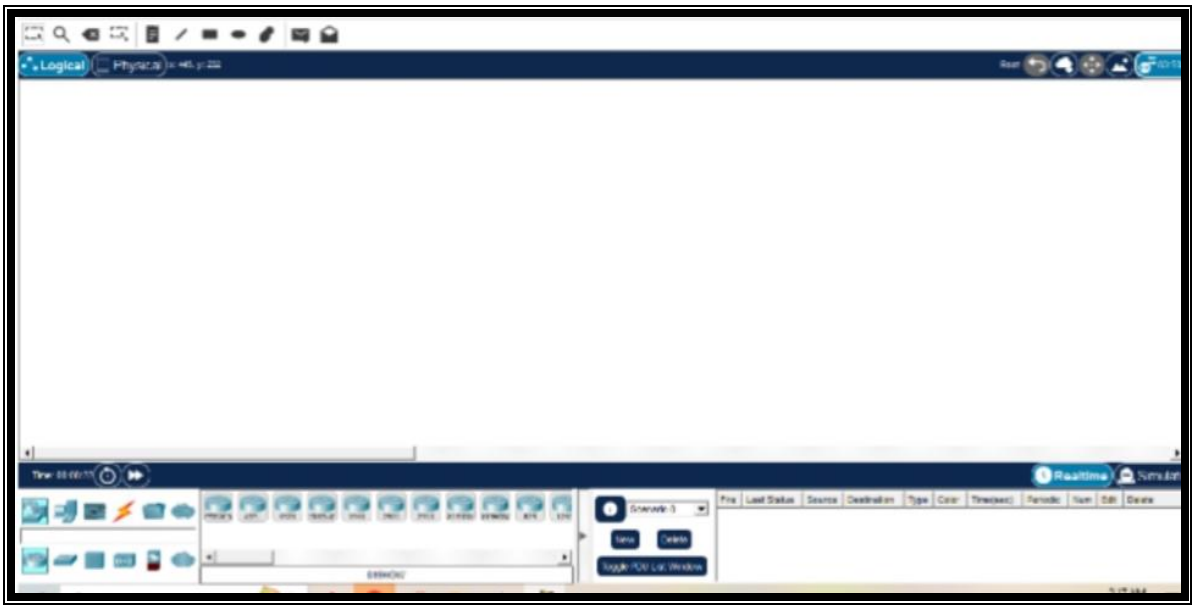
This means communication is successful.

5. Installation of Cisco Packet Tracer




Click Install to continue with the installation, or click Back if you want to review or change any settings.
Destination location:
C:\Program Files\Cisco Packet Tracer 9.0.0
Start Menu folder:
Cisco Packet Tracer
Additional tasks:
Additional shortcuts:
Create a desktop shortcut
Back
Install
Cancel
2/1/2026 4:51 AM
TMP File
499 KB


Networking Academy
Cisco Packet Tracer
LOGIN
Keep me logged in (for 3 months)
Not recommended for public or shared computers
Login Server
Worldwide
Advanced Settings
By using Cisco Packet Tracer you accept Terms of Service (General Terms & SEULA) and the Privacy Statement



Part 2: Explore Packet Tracer Interface

When the software opens you will see:

1. Device Panel

Bottom left area containing:

- Routers
- Switches
- End Devices
- Wireless devices

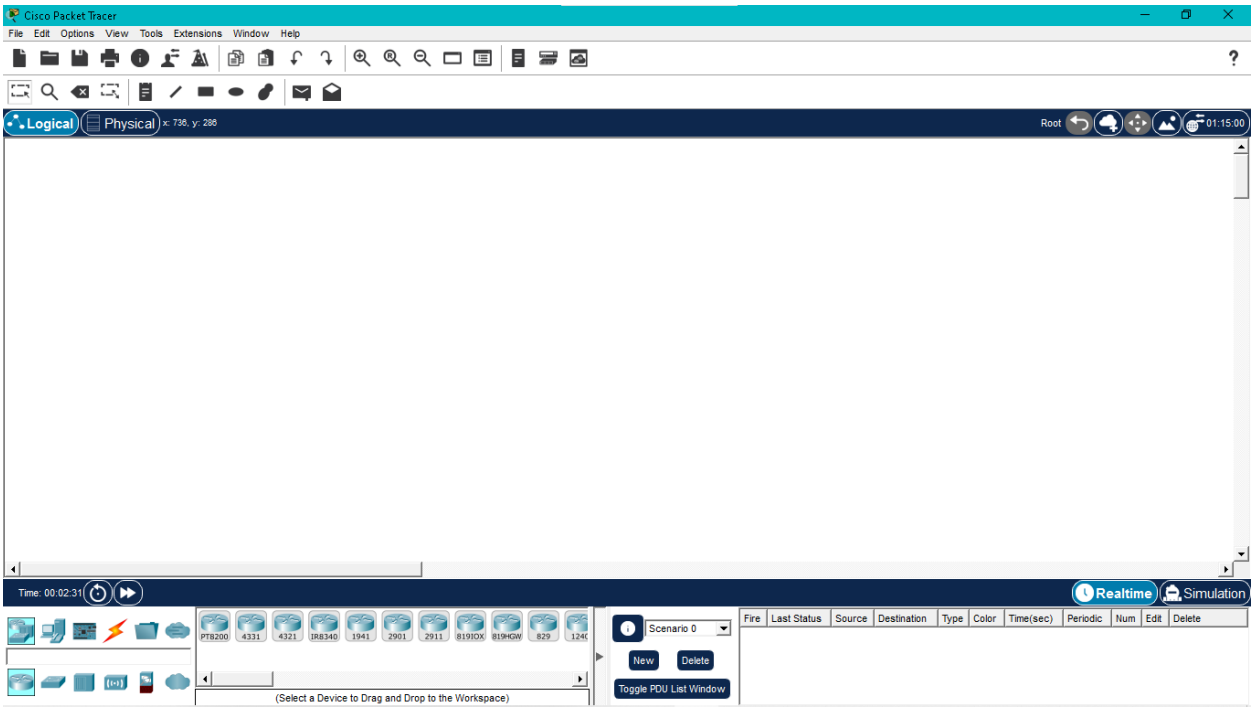
2. Connections Panel

Used to connect devices using cables like:

- Copper Straight-Through
- Copper Cross-Over

3. Workspace

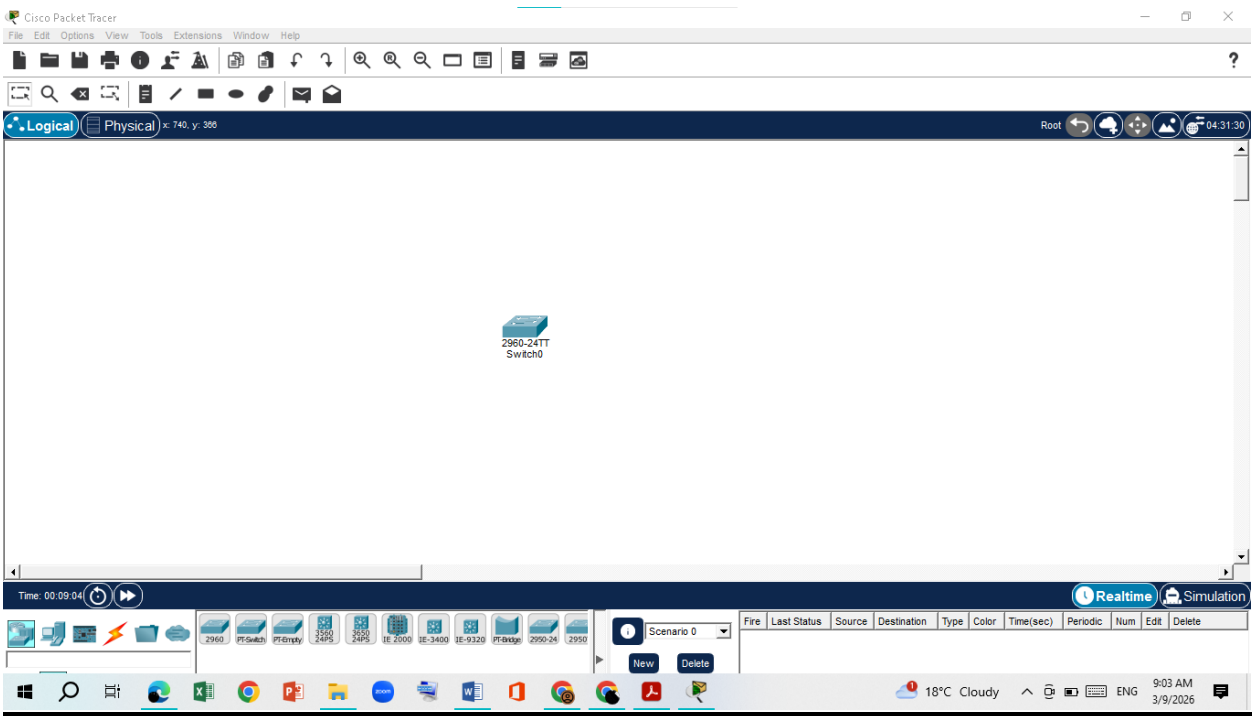
Large white area where network topology is created.



Part 3: Create Network Topology

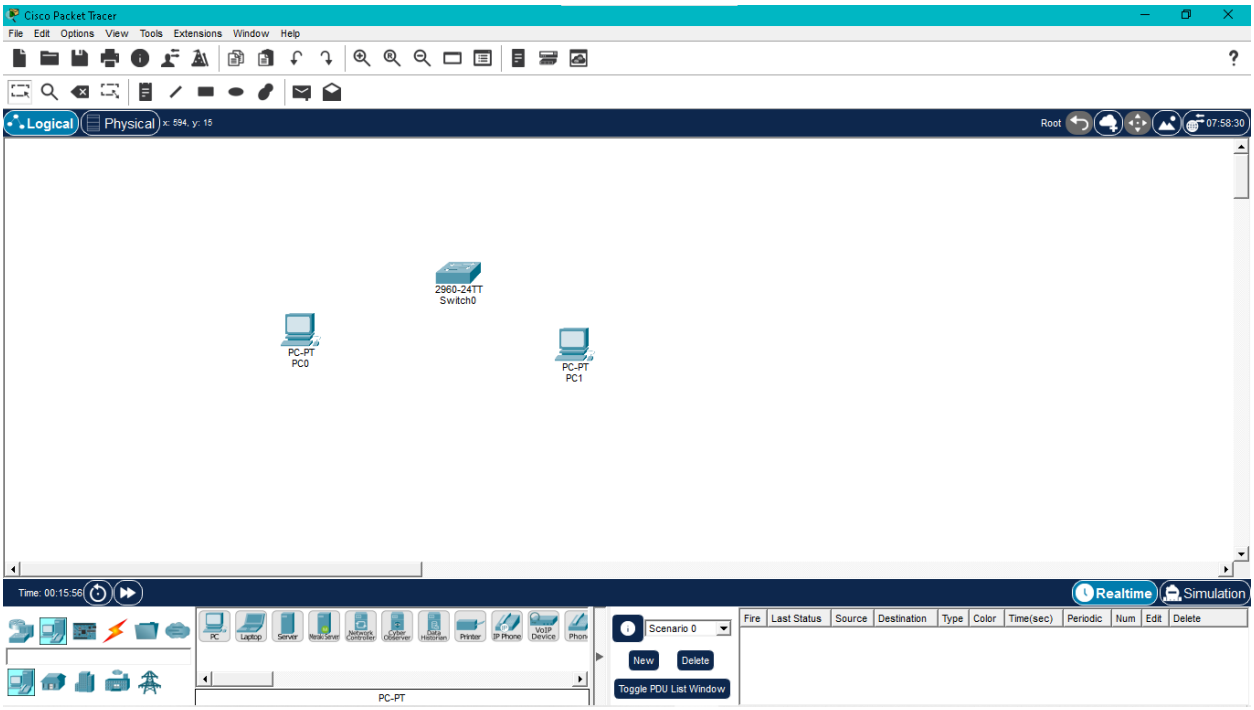
Step1: Adding switch

A switch connects multiple devices in the same network.



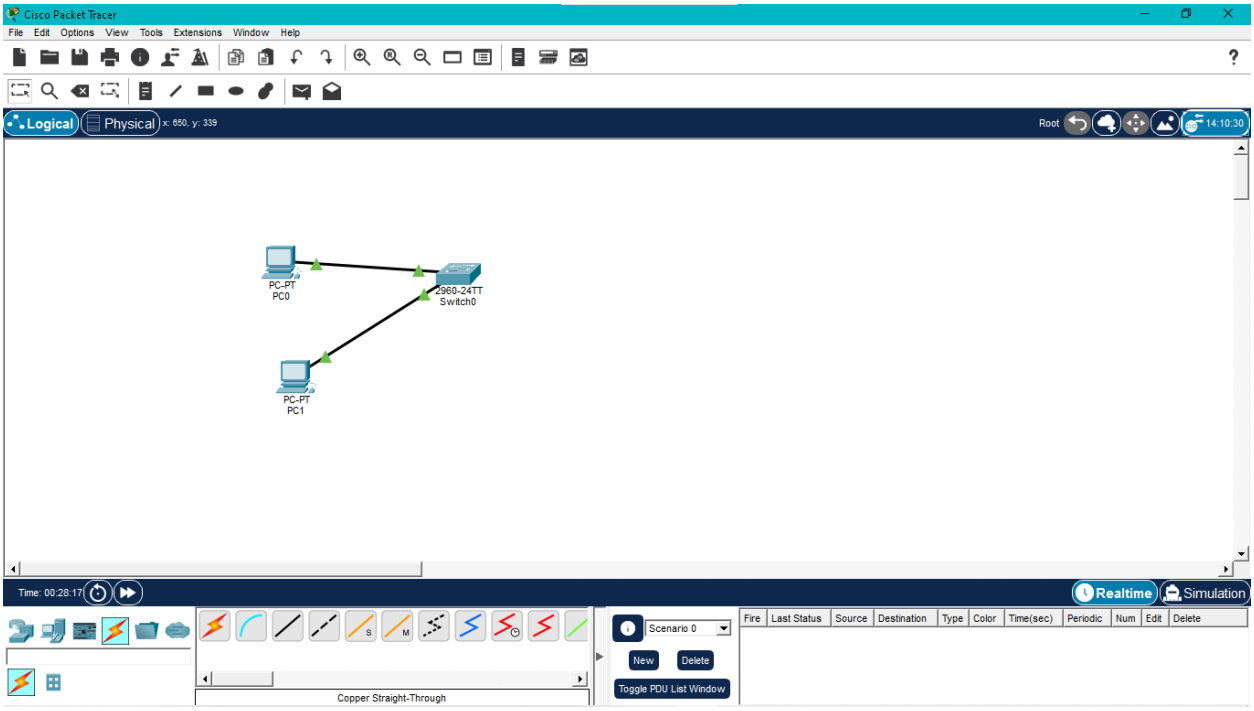
Step 2: Add Two PCs

PCs are end devices used by users to send and receive data.



Part 4: Connect Devices

Copper straight-through cable is used to connect end devices to switches.



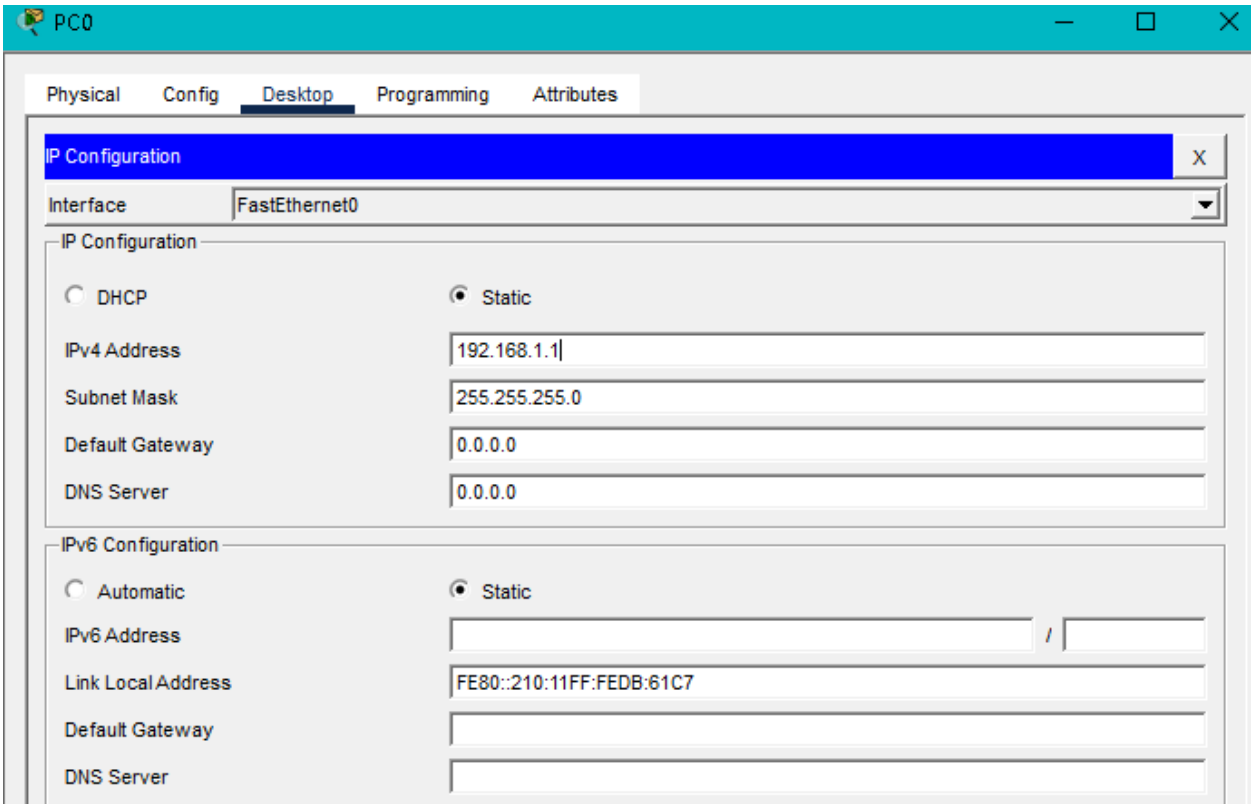
Part 5: Configure IP Address

Each PC needs a **unique IP address**.

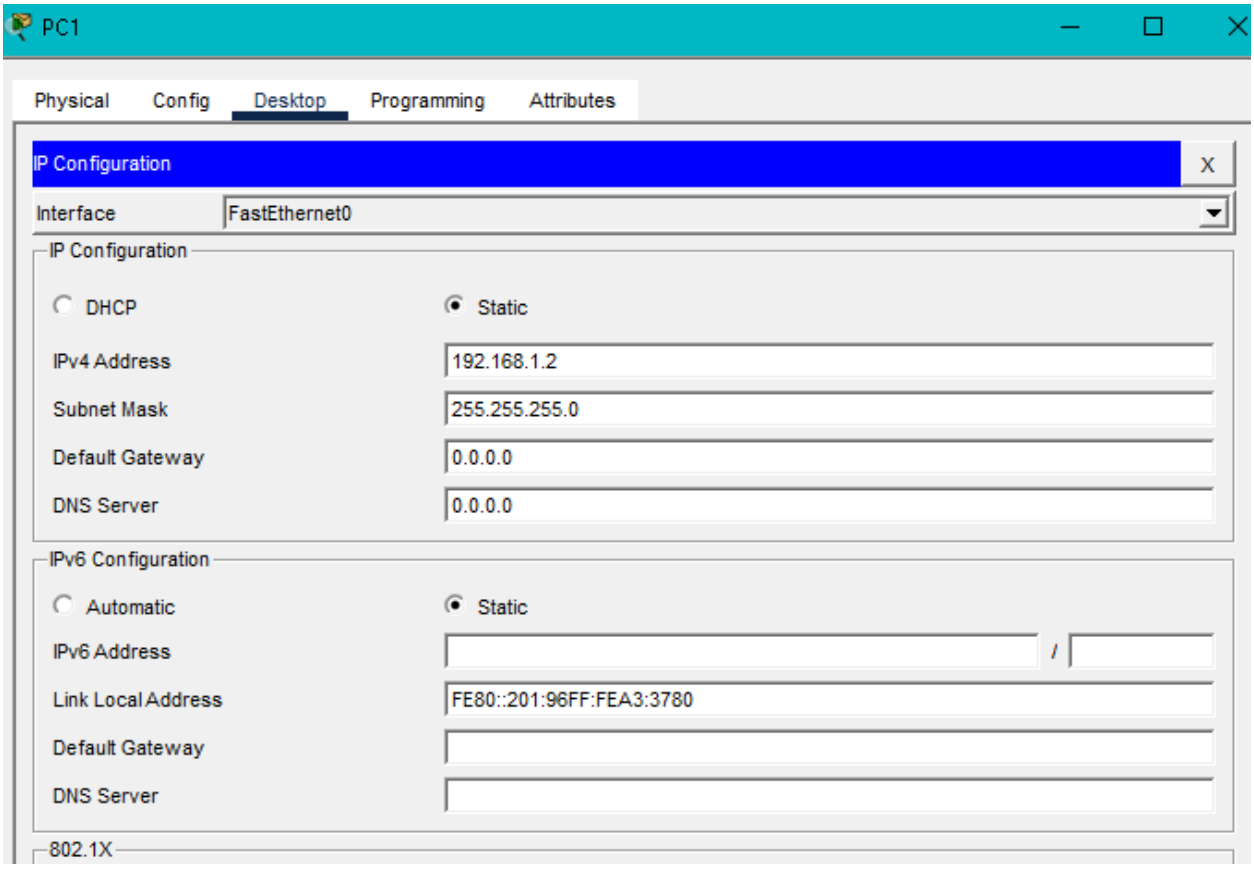
Network Details:

- Network: **192.168.1.0**
- Subnet Mask: **255.255.255.0**

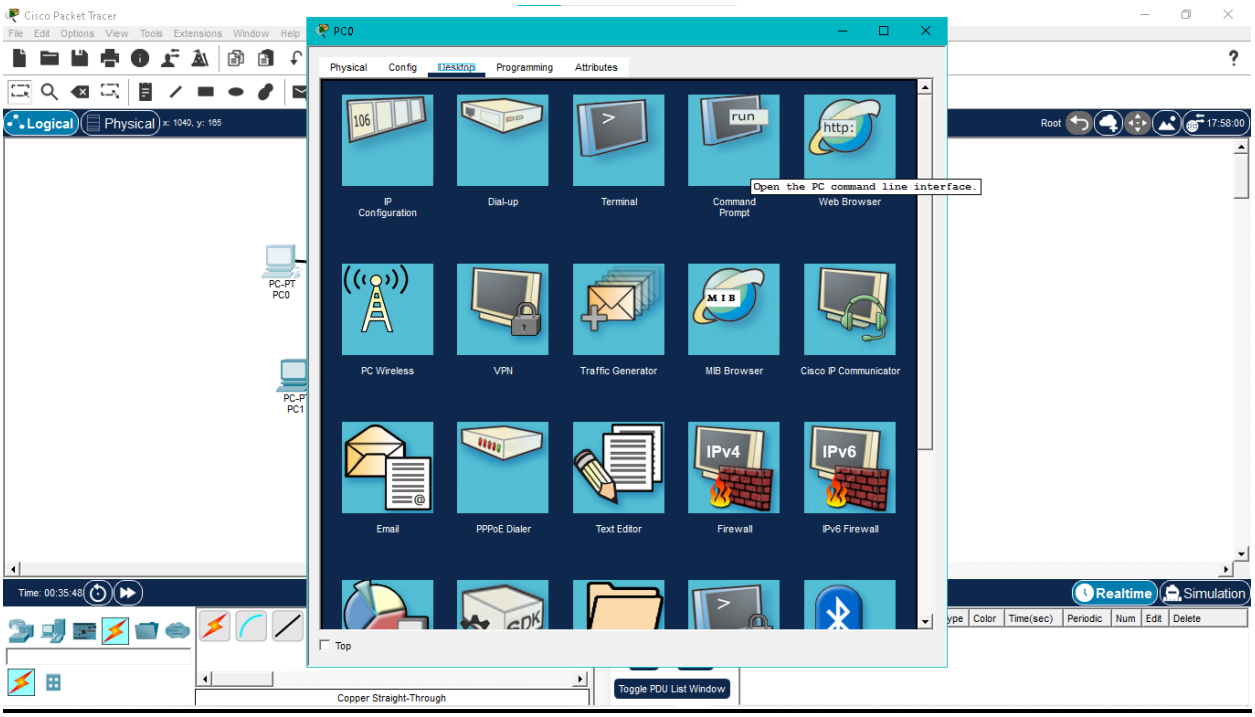
Configure PC0

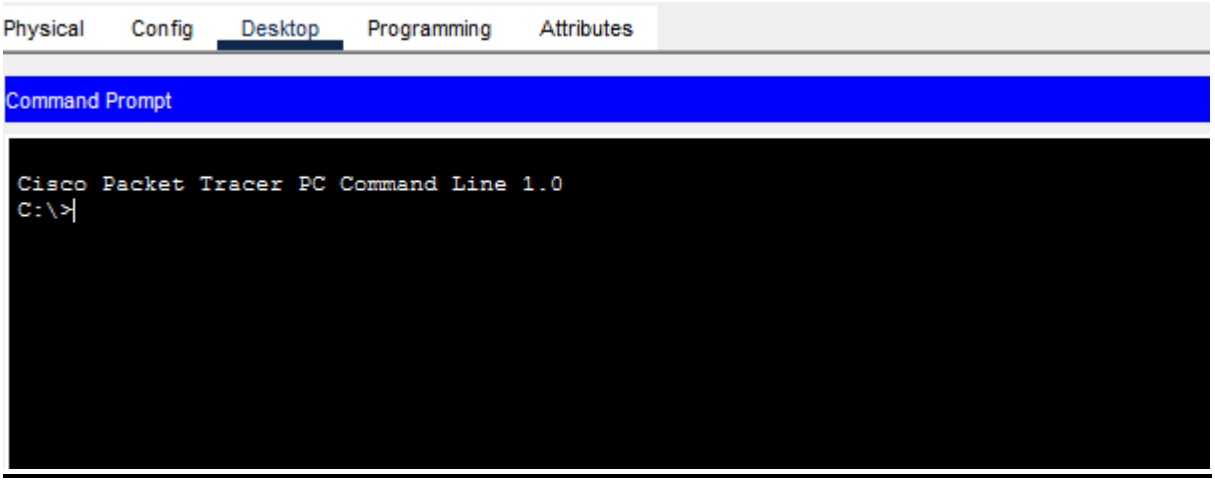


Configure PC1



Part 6: Test Network Using CLI Commands





Command 1: ipconfig

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::210:11FF:FEDB:61C7
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.1
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

Shows IP Address and Subnet Mask.

<u>Command 2: ping</u>

```
                                     0.0.0.0

C:\>ping
Cisco Packet Tracer PC Ping

Usage: ping [-n count | -v TOS | -t ] target

C:\>
```

Command 3: tracert

```
C:\>tracert
Cisco Packet Tracer PC Tracert

Usage: tracert target

C:\>
```

Shows the path packets travel.

Command 4: arp –a

```
C:\>arp -a
No ARP Entries Found
C:\>
```

Shows IP address and MAC address mapping.

Command 5: netstat

```
No ARP Entries Found
C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
C:\>
```